

Dylan Khor

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EDUCATION

Iowa State University

Ames, IA

Master of Science in Computer Science (CGPA: 3.96)

Aug. 2023 – Dec. 2025

- **Thesis:** “Sampling-based Optimized Adaptive Discretization and its Applications in Robotics”

Iowa State University

Ames, IA

Bachelor of Science in Bioinformatics and Computational Biology (CGPA: 3.98)

Aug. 2020 – May 2023

EXPERIENCE

Software Developer

Feb. 2026 – PRESENT

Musco Lighting

Urbandale, IA

- Integrated Pact contract testing into CI/CD pipelines to automate microservice compatibility validation, reducing integration risk and enabling safer deployments.
- Migrated from tunnel-based HTTP request/response communication to an MQTT-based RPC architecture (AWS IoT Core, Eclipse Mosquitto) for cloud-to-edge communication, eliminating tunnel-availability dependencies and improving the reliability, monitorability, and scalability of command delivery.
- Built a containerized Go microservice that ingests AWS IoT Core telemetry via SQS long polling and routes messages to downstream REST services by MQTT topic pattern, with dead-letter-queue handling for non-transient messages.

Apprentice Software Engineer

May 2025 – Aug. 2025

Source Allies

Urbandale, IA

- Deployed IaC using Terraform, integrating Azure Functions and Service Bus for event-driven, asynchronous communication between HR tool and dependent systems, reducing response times.
- Improved system security, reliability, and observability by migrating to OIDC authentication and enhancing monitoring for internal tools.
- Enhanced AI-powered chatbot (FastAPI, React) with multi-session support and retrieval-augmented generation pipeline with GitHub integration, increasing response accuracy and usability.

Graduate Research and Teaching Assistant

Aug. 2023 – Dec. 2025

Iowa State University

Ames, IA

- Researched on adaptive discretization and safety testing techniques in robotics to improve real-world applicability and efficiency.
- Applied expertise in Python programming, algorithm design, and machine learning to mentor students toward academic success.
- Recognized with the Teaching Excellence Award (Fall 2024) for outstanding performance.

PROJECTS

Portfolio Website | *React, Vite, Tailwind CSS, Three.js*

- Developed a personal portfolio optimized for accessibility and responsiveness across devices.
- Integrated interactive 3D visuals with Three.js and GSAP scroll animations for an engaging user experience.

3D Modeler | *C++, Vulkan*

- Built a computer graphics interface for 3D object modeling, compatible with tools like Blender and SolidWorks.

Randomized Progressive Deblurring for Image Classification | *Python, PyTorch, OpenCV*

- Proposed a new training approach for image classification models, achieving a 24% average improvement in generalization.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Go, Java, C++, HTML/CSS, SQL, Bash

Frameworks & Libraries: React, Tailwind CSS, Node.js, pytest, Flask, FastAPI, PyTorch, OpenCV, MuJoCo, Pact

Database: DynamoDB, PostgreSQL, Redis, MongoDB

Cloud & Infrastructure: AWS, Azure (**Fundamentals Certified**), Terraform

Developer Tools: Git, Docker, Linux, Postman, GitHub Actions, GitLab CI